

Learning Outcomes

At the end of the session, you will be able to:

- Write, run, and explain different selection statement and repetition statement.

Activity

1. Write a R program to check whether a year is leap year or not.

Sample expected outcome:

```
Input year: 2004
Output: 2004 is a leap year.

Input year: 1900
Output: 1900 is a not leap year.
```

```
year <- readline(prompt="Input year: ")
year <- as.numeric(year)
if((year%%4==0)&(year%%100!=1)) {
  print(paste(year, "is a leap year"))
} else {
  print(paste(year, "is not a leap year"))
}
```

2. Write a R program to display the cube of the number up to a given integer.

Sample expected outcome:

```
Input an integer: 5
Number is: 1 and cube of the 1 is :1
Number is: 2 and cube of the 2 is :8
Number is: 3 and cube of the 3 is :27
Number is: 4 and cube of the 4 is :64
Number is: 5 and cube of the 5 is :125
```

```
int <- readline(prompt="Input an integer:")
int <- as.numeric(int)
cnt = 1
while (cnt <= int) {
  print(paste("Number is:", cnt, "and cube of the", cnt,"is :",
cnt**3))
  cnt = cnt + 1
}
```

3. Write a R program to check Armstrong number of n digits. An Armstrong number is one whose sum of digits raised to the power three equals the number itself. Example $1^3 + 6^3 + 3^3 + 4^3 = 1634$

Sample expected outcome:

Check whether an n digits number is Armstrong or not:

Input an integer: 1634
1634 is an Armstrong number.

```
print("Check whether an n digits number is Armstrong or not:")
int <- readline(prompt="Input an integer:")
int <- as.numeric(int)

n = 0
arm = int
while (arm != 0) {
  arm = arm%/%10
  n=n+1
}
r=0
result=0
arm = int
while (arm != 0) {
  r = arm%/%10
  result = result + (r**n)
  arm = arm%/%10
}

if(result==int) {
  print(paste(int, "is an Armstrong number. "))
} else {
  print(paste(int, "is an Armstrong number. "))
}
```

Submission

- Submit to your GA.